

B - Inequalities

1(a) $\frac{3-x}{-4} \geq 0$ negative positive
 $\Rightarrow x \in [3, \infty)$

(b) $\frac{-4}{3-x} < 0$ neg pos
 $\Rightarrow x \in (-\infty, 3)$

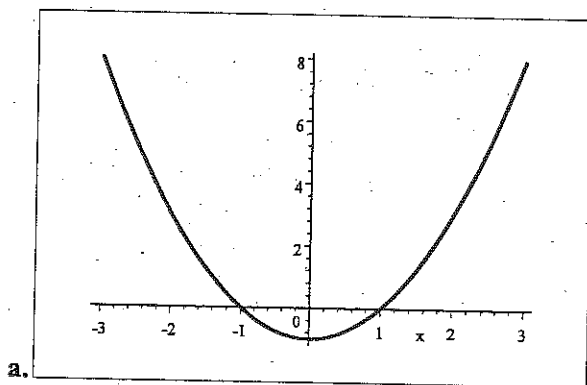
(c) $\frac{2x-6}{3x+9} \geq 0$ pos neg pos
 $\Rightarrow x \in (-\infty, -3) \cup [3, \infty)$

(d) $\frac{x-1}{x} < 4$
 $\Rightarrow \frac{x-1}{x} - 4 < 0$
 $\Rightarrow \frac{x-1-4x}{x} < 0$ neg pos neg
 $\Rightarrow \frac{-3x-1}{x} < 0$
 $\Rightarrow x \in (-\infty, -\frac{1}{3}) \cup (0, \infty)$

(e) $\frac{|-4x|}{3-x} < 0$ pos pos neg
 $\Rightarrow x \in (3, \infty)$

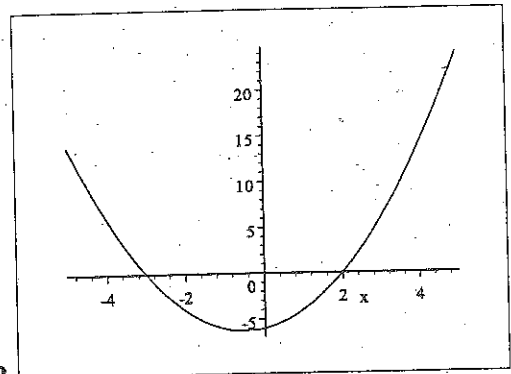
(f) $\frac{2x-6}{13x+9} \geq 0$ neg neg pos
 $\Rightarrow x \in [3, \infty)$

2(a) $x^2+4 > 5 \Rightarrow x^2+4-5 > 0$
 $\Rightarrow x^2-1 > 0 \Rightarrow x \in (-\infty, -1) \cup (1, \infty)$



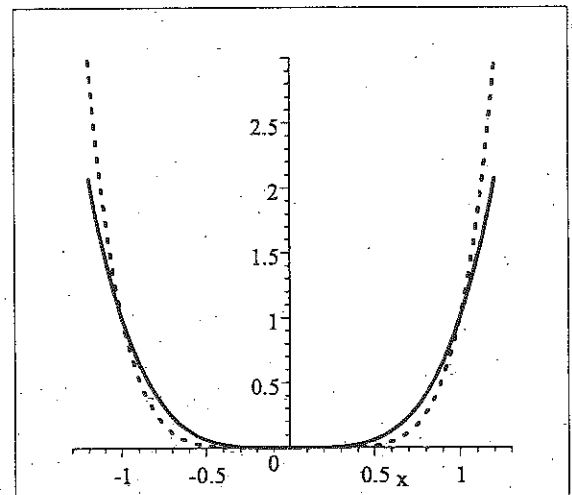
$y = x^2 - 1$

2(b) $x^2-4 > 2-x$
 $\Rightarrow x^2-4-2+x > 0$
 $\Rightarrow x^2+x-6 > 0$
 $\Rightarrow (x+3)(x-2) > 0$
 $\Rightarrow x \in (-\infty, -3) \cup (2, \infty)$



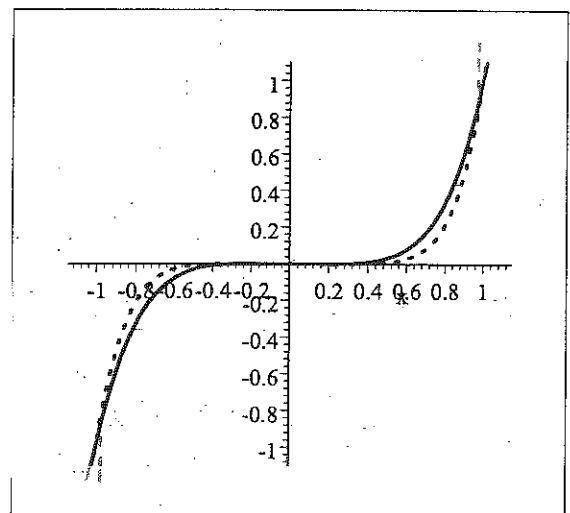
b. $y = x^2 - x - 6$

2(c) $x^6 > x^4$
 $\Rightarrow x \in (-\infty, -1) \cup (1, \infty)$



c. $y = x^6$ is dotted

2(d) $x^5 < x^7$
 $\Rightarrow x \in (-1, 0) \cup (1, \infty)$



$y = x^7$ is dotted